
Hedge Fund Trading and Sovereign Bond Yield Sensitivity

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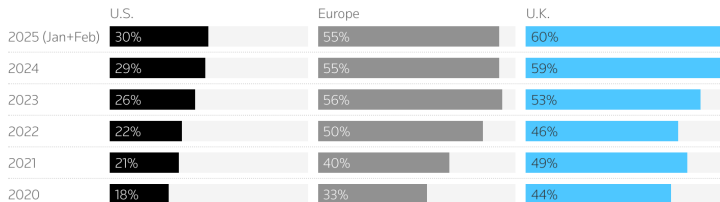
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Hedge funds hold a growing footprint in government bond markets

Hedge funds in govies

Notional percentage volumes in government bonds tracked by Tradeweb

● U.S. ● Europe ● U.K.



Source: Tradeweb | Chart by Nell Mackenzie

- Their share of trading volume keeps rising since 2020
- In Europe it went from a third to more than half of volumes
- The rise is broad, across the US, Europe, and the UK

► Do hedge funds change how prices incorporate new information?

Two views on the impact of leveraged investors on price

*“On the one hand, hedge funds are frequently credited with **enhancing market efficiency and liquidity**. On the other hand, they could also act as potential **amplifiers of volatility and market shocks**.”*

The ECB Blog, *Hedge funds: good or bad for market functioning?*, September 2024

Decrease sensitivity

Their arbitrage activity corrects mispricings
Prices are pushed *toward* fundamentals

Gromb and Vayanos (2002) and Cao et al. (2018)

Increase sensitivity

Leverage and funding fragility force trades
Prices are pushed *away* from fundamentals

Shleifer and Vishny (1997), Brunnermeier and Pedersen (2009), and
Geanakoplos (2010)

Whether hedge funds increase or decrease price sensitivity in a given situation is ultimately an empirical question, and granular evidence remains scarce.

► **This paper shows hedge funds amplify the sensitivity of sovereign bond prices to shocks**

The financial system has shifted toward nonbank intermediation

- Tighter bank regulation since the global financial crisis moved activity toward nonbank financial intermediaries (Pelizzon et al. [2025a](#))
- Offshore hedge funds stand out for their **opacity** and heavy use of **leverage** (Barth et al. [2025](#); Aramonte et al. [2023](#))
- They finance positions with short term repo, which makes them sensitive to sudden moves in collateral value

The risk is procyclical amplification

- Leverage can lead funds to trade with the shock rather than against it, amplifying the move instead of absorbing it
- March 2020 was an extreme example, when forced deleveraging by hedge funds intensified stress in the US Treasury market (Barth and Kahn [2025](#); Kruttli et al. [2025](#))

What we still do not know

- Whether this trading shapes pricing **outside crises**, and whether assessing the risk needs more than the presence and size of positions

Research question

- Does the **leveraged positioning** of hedge funds amplify sovereign bond yield sensitivity to **shocks**, and under what conditions?

Taking the question to the data

1. Leveraged positioning

Constructed from regulatory repo data that track daily hedge fund positions in individual sovereign bonds

2. Shocks

Measured with high frequency monetary policy surprises that move yields exogenously (Altavilla et al. [2019](#))

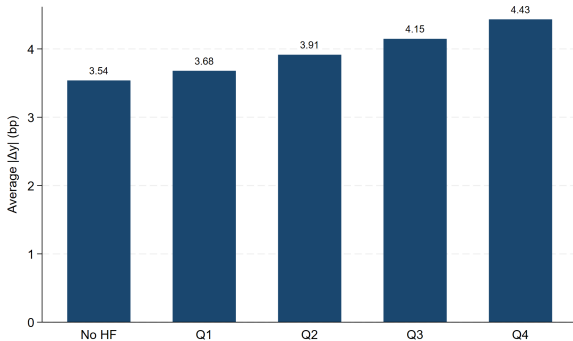
Strategy

- Interact **predetermined** hedge fund positions with the shock to recover the additional yield sensitivity that arises when hedge funds trade a given bond

Four findings

1. Hedge funds increase the volatility of bond yields on shock days
2. Hedge fund trading raises the sensitivity of yields to shocks
3. The effect is symmetric, whether the shock helps or hurts the position
4. It is concentrated in directional books

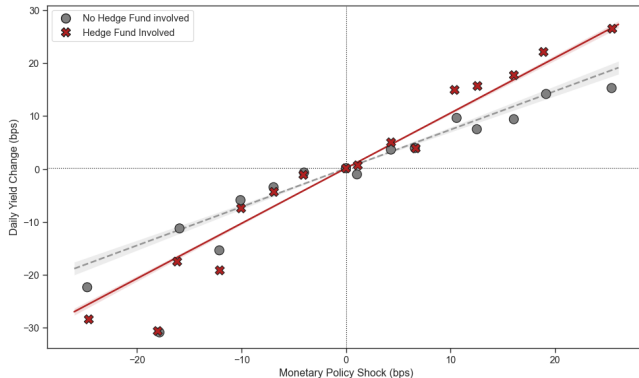
Finding 1: Hedge funds increase the volatility of bond yields on shock days



Average absolute daily yield change in basis points on days without a monetary policy shock. Buckets are quartiles of hedge fund intensity, the five day net position scaled by amount outstanding, among bonds with active positions, next to a no hedge fund reference. Raw group means without controls. Sample period 2021 to 2025.

- Hedge funds select more volatile bonds
- Excess volatility is limited to shock days

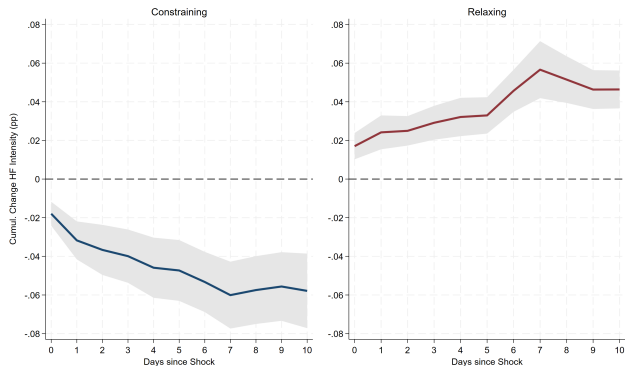
Finding 2: Hedge fund trading raises the sensitivity of yields to shocks



Binned scatterplot of daily yield changes against monetary policy shocks, separately for bonds with and without active hedge fund positions. Yield changes are residualized for bond fixed effects. Lines are linear fits on the bond level data. Sample period 2021 to 2025.

► Not price discovery, yields overshoot and revert within days, a temporary dislocation

Finding 3: The effect is symmetric

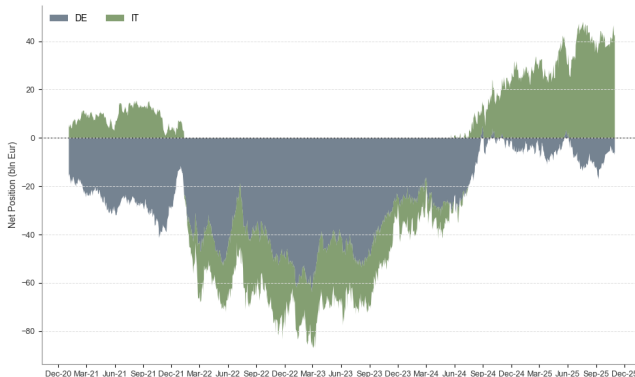


Cumulative change in hedge fund intensity from $t - 1$ to $t + h$ after a shock, by regime, local projections on overnight repo (Jordà 2005). Bond and duration $\times$ country $\times$ date fixed effects. Shaded areas are 90% confidence intervals. Sample period 2021 to 2025.

► Consistent with rebalancing toward a target

Term repo

Finding 4: It is concentrated in directional books



Aggregate net repo positions of offshore hedge funds in German and Italian sovereign bonds, EUR outstanding. Positive values are net cash borrowing (a proxy for long), negative values net cash lending (a proxy for short). Sample period 2021 to 2025.

► So the directionality of the book matters, beyond how much hedge funds hold

- **Determinants of sovereign yields and non fundamental pricing**

I document a specific source of temporary deviations from fundamentals

De Grauwe and Ji ([2013](#)); Greenwood and Vayanos ([2014](#)); Kojien et al. ([2021](#)); Duffie ([2010](#))

- **Nonbanks and hedge funds in financial markets**

I show an amplification channel that is not confined to stress episodes

Barth and Kahn ([2025](#)); Kruttli et al. ([2025](#)); Anaya et al. ([2025](#)); Pelizzon et al. ([2025b](#))

- **How leveraged investors transmit shocks**

I show comparable amplification in the constraining and the relaxing state

Adrian and Shin ([2014](#)); Brunnermeier and Pedersen ([2009](#)); Avalos and Sushko ([2023](#)); De Long et al. ([1990](#)); Moskowitz et al. ([2012](#))

Data

Hedge fund repo positions ECB SFTDS

- Regulatory transaction data covering euro area repo activity at the transaction level
- Offshore hedge funds identified by Cayman Islands domicile, refined with the classification of Lenoci and Letizia ([2021](#)), giving **152 funds**

Bond characteristics Bloomberg

- Daily yields, duration, bid ask spread, amount issued
- Cheapest to deliver status built from Eurex delivery outcomes ([Pelizzon et al. 2025b](#))

Exogenous shocks EA-MPD ([Altavilla et al. 2019](#))

- Intraday change in the 2 year OIS rate during the ECB monetary event window
- The same shock hits every bond, so identification comes from cross sectional differences in positioning

Sample

- German and Italian sovereign bonds, Jan 2021 to Nov 2025 (about 70% of euro area hedge fund sovereign activity)

Hedge funds hold large, leveraged positions

	Mean	Median
<i>Footprint</i>		
HF present (share of bond days)	0.53	1.00
Net position when active (% issued)	3.01	1.80
<i>Leverage and funding</i>		
Haircut, long (%)	0.24	0.12
Haircut, short (%)	-0.08	-0.05
Maturity, long (days)	16.5	11.0
<i>Shock days</i>		
Tightening surprise (bps)	5.87	4.95
Easing surprise (bps)	-4.17	-3.13

Daily panel of German and Italian sovereign bonds. Footprint figures are conditional on active hedge fund participation. Haircuts and maturities use active transactions. Sample period 2021 to 2025.

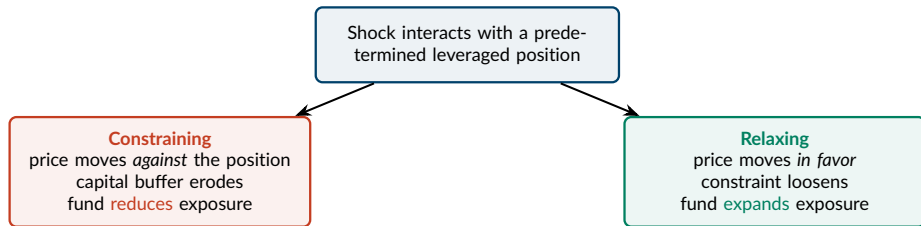
- Hedge funds are active on **53%** of bond days
- When active they hold **3%** of the bond on average, an economically meaningful share
- Haircuts near **zero** implying high potential levels of leverage on these safe assets
- A typical surprise of about **4–6 bps** is enough to move a position materially

Mechanism

How leveraged positioning amplifies yields

Starting point Adrian and Shin (2014)

- Leveraged intermediaries actively manage their balance sheet against the risk environment
- The same rule produces **opposite** position adjustments depending on the sign of the shock



► In both states the resulting trading pushes yields further in the direction of the initial shock

A numerical example

Setup representative sample values

- A bond of duration about 7 years financed via repo, effective margin $m_i = 2\%$ (raw haircuts 0.24%)
- Hedge funds hold $Q_{HF} / Q_{Total} = 3\%$ of the bond
- Tightening surprise of 6 bps, market elasticity $\lambda = 5$ bps per pp traded (D'Amico and King [2013](#); Eser and Schwaab [2016](#); De Santis and Holm-Hadulla [2020](#))

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Step 1 the shock erodes the capital buffer

$$\frac{\Delta E}{E} \approx \frac{1}{m_i} \times \frac{\Delta P}{P} = \frac{-0.42\%}{2\%} = -21\%$$

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Step 3 trading pressure moves the yield

$$\Delta y^{\text{Amp}} = \lambda \times 0.63 = 5 \times 0.63 = 3.15 \text{ bps}$$

Empirical Strategy

Baseline panel regression

$$\Delta y_{i,t} = \beta_1 (\text{HF}_{i,t-1} \times \text{Shock}_t) + \beta_2 \text{HF}_{i,t-1} + \beta_3 \text{Shock}_t + \gamma X_{i,t-1} + \theta_k + \varepsilon_{i,t}$$

Two measures of positioning

- **Presence**, a binary indicator of an active position
- **Intensity**, the five day net position scaled by amount outstanding

Coefficient of interest β_1

- Additional yield sensitivity to the shock when hedge funds are active

Two challenges to causal reading

1. Hedge funds **select** certain bonds
2. The effect could reflect **dealer** behavior

Hedge funds do not pick bonds at random

HF involvement	(1)	(2)	(3)
Duration	0.012***	0.012***	0.013***
Bid ask spread	-0.534***	-0.524***	-0.549***
Amount issued	0.030***	0.032***	0.032***
CTD flag	0.070**	0.070**	0.070**
Country FE	No	Yes	No
Country $\times$ Date FE	No	No	Yes
R^2	0.518	0.533	0.543

Linear probability model, dependent variable is a binary indicator for offshore hedge fund repo activity. $N = 477,001$. Standard errors clustered by date and bond. *** $p < 0.01$, ** $p < 0.05$.

Hedge funds tilt toward **liquid**, **larger**, longer duration, and cheapest to deliver bonds, which could move shock sensitivity on their own.

► **A simple correlation would confound leveraged capital with the bonds hedge funds prefer**

The main specification compares similar bonds on the same day

Duration $\times$ country $\times$ date fixed effects

- The effect is identified within cells of bonds with similar duration from the same issuer on the same day
- Bond level controls absorb remaining differences in liquidity and size within the cell

A matched design additionally makes the comparison group explicit

- Each hedge fund bond is paired with the non hedge fund bond in its cell closest in bid ask spread, estimated with pair fixed effects
- No functional form imposed on how characteristics map into sensitivity, each bond is benchmarked only against its closest twin

► The bond types hedge funds select no longer drive the comparison

Results

Finding 1

Hedge funds increase the volatility of bond yields on shock days

Comparable bonds are more volatile only on shock days

$ \Delta \text{yield} $ (bps)	(1)	(2)	(3)	(4)
HF involved $\times$ Shock	0.240***	0.241***	0.253***	0.246***
Shock	0.545***	0.541***		
HF involved (level)	0.459***	0.817***	-0.128*	-0.098
Bond controls	No	Yes	Yes	Yes
Bond FE	No	No	Yes	Yes
Date FE	No	No	Yes	No
Duration $\times$ Date $\times$ Country FE	No	No	No	Yes

Dependent variable is the absolute daily yield change. Controls included from column 2. Standard errors clustered by date and bond.

*** $p < 0.01$, * $p < 0.10$.

The **level** differential disappears once we compare bonds of similar duration from the same issuer on the same day (column 4).

The **interaction** with the shock stays positive throughout. Extra volatility appears only when a shock meets the position.

► For that reason the preferred design uses duration $\times$ country $\times$ date fixed effects

Finding 2

Hedge fund trading raises the sensitivity of yields to shocks

Hedge fund presence amplifies the yield reaction

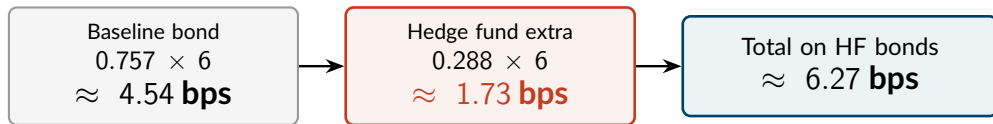
Δ yield (bps)	(1)	(2)	(3)	(4)	(5)
HF involved $\times$ Shock	0.288***	0.288***	0.287***	0.291***	0.304***
Shock	0.757***	0.757***			
HF involved (level)	0.083**	0.062	−0.000	−0.022	−0.002
Bond controls	No	Yes	Yes	Yes	Yes
Bond FE	No	No	Yes	Yes	No
Date FE	No	No	Yes	No	No
Dur. $\times$ Date $\times$ Country FE	No	No	No	Yes	No
Matched pair FE	No	No	No	No	Yes

Dependent variable is the daily yield change. Column 5 is the matched sample. Standard errors clustered by date and bond, and by cell and twin in column 5. *** $p < 0.01$, ** $p < 0.05$.

The interaction is stable from 0.288 to 0.304 across progressively tighter designs, while the direct effect of presence stays at zero.

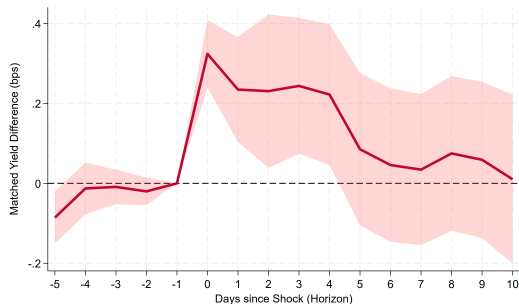
The amplification is economically large

For an average tightening surprise of 6 bps, using column 1.



► More than a quarter of the yield move on shock days in these bonds is associated with the trading of hedge funds, not with the fundamental news

Yields diverge after a shock

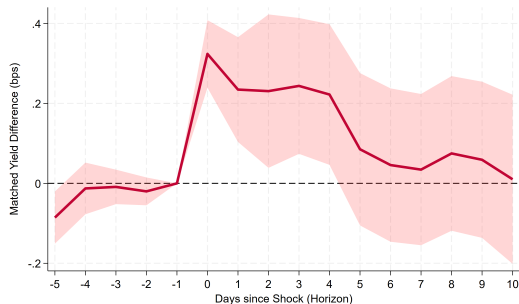


(a) Matched differential

Local projections on the matched sample (Jordà 2005). Shaded areas are 90% confidence intervals clustered by date and bond.

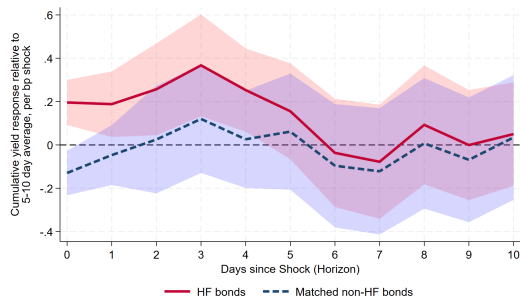
Flat before the shock, immediate divergence on impact,
significant for about **four days**, then reversion.

Yields diverge after a shock, reflecting temporary overshooting



(a) Matched differential

Local projections on the matched sample (Jordà 2005). Shaded areas are 90% confidence intervals clustered by date and bond.



(b) Level decomposition

Flat before the shock, immediate divergence on impact, significant for about **four days**, then reversion.

► **A temporary dislocation that other capital gradually absorbs, consistent with slow moving capital**

Both legs converge to a common level. Hedge fund bonds **overshoot** first.

Not price discovery

The trigger is hedge fund positioning, not dealer behavior

Δ yield (bps)	(1)	(2)	(3)	(4)
HF involved $\times$ Shock	0.271***	0.294***	0.248***	0.250***
HF involved (level)	0.006	0.007*	0.001	0.001
HF $\times$ Shock $\times$ Δ CDS				0.000
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dealer $\times$ Date FE	Yes	No	No	No
Dealer $\times$ Date $\times$ Country FE	No	Yes	No	No
Dealer $\times$ Date $\times$ Dur. FE	No	No	Yes	Yes

Unit of observation is dealer, bond, day, balanced with zeros. Δ CDS is the dealer's own five day CDS change before the shock. Standard errors clustered by dealer, date, and bond.

Comparing bonds traded by the **same dealer on the same day**, the hedge fund bonds still react more. The effect does **not** scale with the dealer's own credit stress.

► **Dealers intermediate the flow, but the trigger is the leveraged positioning of the funds**

The amplification scales with position size

Δ yield (bps)	(1)	(2)	(3)	(4)
HF intensity $\times$ Shock	0.166***	0.165***	0.167***	0.164***
Shock	0.811***	0.811***		
HF intensity (level)	0.008	−0.020	−0.009	−0.011
Bond controls	No	Yes	Yes	Yes
Bond FE	No	No	Yes	Yes
Date FE	No	No	Yes	No
Dur. $\times$ Date $\times$ Country FE	No	No	No	Yes

HF intensity is the log of one plus the five day net position scaled by amount outstanding. Standard errors clustered by date and bond.

*** $p < 0.01$.

Within the **same bond** over time, a larger position produces a larger reaction, which pure selection cannot explain. At a 6 bps shock and 3% intensity this adds about **1.36 bps**.

► **A continuous dose response that complements the matched sample evidence**

Finding 3

The amplification is symmetric across regimes

The amplification holds whether shocks help or hurt the position

Δ yield (bps)	Constraining	Relaxing
HF int. $\times$ Shock	0.140***	0.224***
HF int. (level)	-0.016	-0.009
Bond controls	Yes	Yes
Bond FE	Yes	Yes
Dur. $\times$ Date $\times$ Ctry FE	Yes	Yes

Each regime also includes all non shock days and zero intensity days as the within bond baseline. Standard errors clustered by date and bond.

- **Constraining** when the shock moves against the position
- **Relaxing** when it moves in favor
- A Wald test **cannot distinguish** the two ($p = 0.16$)

Comparable amplification in both states, which is what *rebalancing toward a target* predicts

► Amplification is not only a crisis phenomenon, it operates whenever shocks meet positions

Finding 4

The directionality of the book governs the strength

The constraint binds at the level of the whole book

- A book long Italy and short Germany of similar duration carries little net exposure to a parallel rate move, so a shock barely moves its capital
- A one sided directional book absorbs the full loss and rebalances

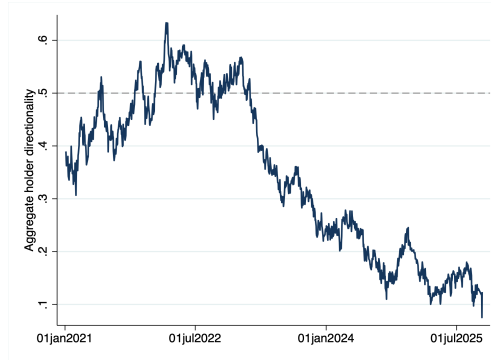
A bond level measure of holder directionality

$$\text{Dir}_{f,t} = \frac{|\sum_i D_i q_{f,i,t}|}{\sum_i D_i |q_{f,i,t}|}, \quad \text{HolderDir}_{i,t} = \frac{\sum_f |q_{f,i,t}| \text{Dir}_{f,t}}{\sum_f |q_{f,i,t}|}$$

Zero when long and short legs offset in duration, close to one when the book is one sided. Each bond inherits the position weighted directionality of the funds holding it.

► **A more hedged book behaves like a higher effective margin in the mechanism**

The sector rotates between hedged and directional books



Position weighted average of holder directionality across bonds. High when bonds are held mainly within directional books. Within duration, country, and date cells the average dispersion is about 0.11, which is the variation that identifies the next result. Sample period 2021 to 2025.

► Directional through the hiking cycle, then back toward hedged relative value in 2024 and 2025

Amplification is concentrated in directional books

Δ yield (bps), relative to no HF	Coefficient
Hedged book $\times$ Shock	0.098***
Directional book $\times$ Shock	0.363***
<i>By directionality quartile</i>	
Q1 most hedged $\times$ Shock	0.057
Q2 $\times$ Shock	0.137***
Q3 $\times$ Shock	0.361***
Q4 most directional $\times$ Shock	0.367***
Bond controls	Yes
Bond FE	Yes
Dur. $\times$ Date $\times$ Country FE	Yes

Bond and duration $\times$ country $\times$ date fixed effects with controls. Wald tests reject equality of the hedged and directional groups ($F = 24.8$) and of the extreme quartiles ($F = 28.3$). Standard errors clustered by date and bond.

The amplification rises **monotonically** with directionality and is **not distinguishable from zero** in the most hedged quartile.

► Same shock, same day, same maturity and country, the directional holders carry the effect

Conclusion

What the paper shows

1. Hedge funds select more volatile bonds, but the excess volatility is limited to shock days
2. Their trading raises the sensitivity of yields to shocks, more than a quarter of the shock day move, and the overshoot reverts within days
3. The amplification is symmetric across the constraining and relaxing states, consistent with constant leverage targeting
4. It is concentrated in **directional** books and close to zero in **hedged** books

For risk monitoring

- Size and presence alone are not sufficient, and amplification is not confined to stress episodes
- The **directionality** of the book is a key risk metric beyond position size

Thank you

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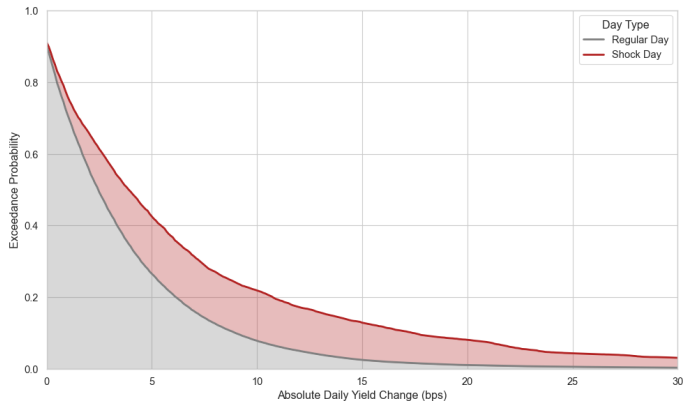
Annex

Summary statistics

	Mean	SD	P25	Median	P75	N
<i>Bond market</i>						
Daily yield change (bps)	0.15	5.79	−2.25	0.00	2.75	477,001
Duration (years)	7.13	7.00	1.77	5.14	10.47	477,001
Bid ask spread (€)	0.31	0.42	0.03	0.11	0.44	477,001
<i>Hedge fund positioning</i>						
HF present (dummy)	0.53	0.50	0.00	1.00	1.00	477,001
Net position, active (% issued)	3.01	3.29	0.64	1.80	4.26	253,111
<i>Repo terms, active</i>						
Haircut, long (%)	0.24	0.54	0.00	0.12	0.30	109,068
Maturity, long (days)	16.49	16.42	7.50	11.00	18.65	101,535
<i>Shock days</i>						
Tightening surprise (bps)	5.87	4.10	3.30	4.95	7.60	5,736
Easing surprise (bps)	−4.17	4.48	−6.72	−3.13	−0.56	8,803

Daily panel of German and Italian sovereign bonds. Positioning variables are five day moving averages scaled by amount outstanding. Sample period 2021 to 2025.

Monetary surprises generate fat tails in yield changes



Share of bond days where the absolute yield change exceeds the threshold on the horizontal axis. Red is monetary announcement days, grey is regular days. The gap is the excess tail risk from surprises. Sample period 2021 to 2025.

► Surprises move yields enough to activate the mechanism

Back

The amplification is not price discovery

A price discovery story predicts **stronger** amplification when the surprise carries new fundamental information.

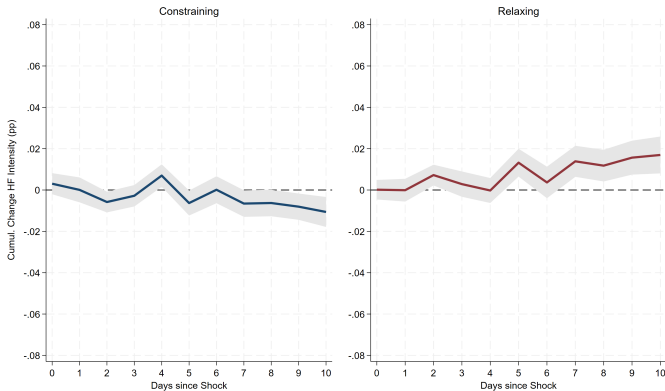
Δ yield (bps)	(1) Pure monetary	(2) Information
HF involved $\times$ Shock	0.377***	0.113***
HF involved (level)	-0.019	-0.026
Bond controls	Yes	Yes
Bond FE	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes

Both columns include bond and duration $\times$ country $\times$ date fixed effects and controls. A Wald test rejects equality ($p = 0.065$). Standard errors clustered by date and bond.

The amplification is concentrated in the **pure monetary** component, the part that carries **no** new fundamental information. This is the **opposite** of the price discovery prediction.

► **Decomposing the surprise points the same way as the overshooting** (Jarociński and Karadi 2020)

Term repo shows the same pattern with a lag



Cumulative change in hedge fund intensity by regime, term repo only, local projections (Jordà 2005). Same contraction and expansion as overnight, delayed by about five days in line with the maturity profile. Sample period 2021 to 2025.

External validity, expanded sovereign sample

Δ yield (bps)	(1) Presence	(2) Intensity	(3) Constraining	(4) Relaxing
HF involved $\times$ Shock	0.192***			
Log HF intensity $\times$ Shock		0.122***	0.107***	0.171***
HF involved (level)	-0.046			
Log HF intensity (level)		-0.015	-0.019	-0.012
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Adds French and Spanish sovereigns to the German and Italian baseline. Bond and duration $\times$ country $\times$ date fixed effects with controls. Standard errors clustered by date and bond. *** $p < 0.01$.

Germany, Italy, France, and Spain. All coefficients stay positive and significant, somewhat smaller, consistent with hedge funds being less active in French and Spanish bonds.

Amplification under a sovereign credit shock

Δ yield (bps)	(1)	(2)	(3) Constraining	(4) Relaxing
HF involved $\times$ Shock	0.225***			
Log HF intensity $\times$ Shock		0.107***	0.129***	0.117***
HF involved (level)	−0.030			
Log HF intensity (level)		−0.021	−0.022	−0.020
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Standard errors clustered by date and bond. *** $p < 0.01$.

The shock is the residual of the daily change in the five year US dollar Italian sovereign CDS spread, restricted to its 2% tails, away from ECB events. The two regimes are close in magnitude.

Volatility standardized yield changes

Standardized Δ yield (σ units)	(1)	(2)	(3) Constraining	(4) Relaxing
HF involved $\times$ Shock	0.043***			
Log HF intensity $\times$ Shock		0.020***	0.016***	0.031***
HF involved (level)	−0.007			
Log HF intensity (level)		−0.004	−0.005	−0.004
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Standard errors clustered by date and bond. *** $p < 0.01$.

Each yield change is divided by the bond's own conditional volatility from a GARCH(1,1) fit on non shock days. With average $\hat{\sigma}$ near 6 bps these map back to roughly 0.26 and 0.12 bps per bp, close to the baseline.

Placebo with shifted shock dates

Δ yield (bps)	(1)	(2)	(3) Constraining	(4) Relaxing
HF involved $\times$ Placebo shock	−0.030			
Log HF intensity $\times$ Placebo shock		−0.018	−0.014	−0.028
Placebo shock	0.007	0.004	0.002	−0.003
HF involved (level)	−0.031			
Log HF intensity (level)		−0.018	−0.017	−0.020
Bond controls	Yes	Yes	Yes	Yes
Bond FE	Yes	Yes	Yes	Yes
Dur. $\times$ Date $\times$ Country FE	Yes	Yes	Yes	Yes

Monetary shock dates shifted by 15 trading days, which breaks the timing. Standard errors clustered by date and bond.

Shifting the shock dates removes the coincidence with announcements. The amplification disappears entirely, which ties it to the actual shock.